

# Advance Java Topics or Syllabus

## 1. Basics of a Web application

- What is a web application?
- What is a web client and web server?
- How do client and server communicate?
- HTTP protocol basics
- HTML language basics
- What is a TCP/IP port, URL?
- Need for a Web Container

## 2. Web Container and Web Application Project Set up

- To set up Tomcat Container on a machine
- To set up a Servlets JSP project in Eclipse
- To configure dependency of Servlet JSP APIs
- Web application project structure

## 3. Servlets

- What are Servlets?
- What can they do? Why are they needed?
- How do Servlets look in code?
- HTTP Methods; GET, POST, PUT, DELETE, TRACE, OPTIONS
- GET/POST request; differences between the two
- Servlet Lifecycle
- Servlet Context and Servlet Config
- Forwarding and Redirection of requests

## 4. Session Management

- What is a session?
- Why is it required?
- How to get a session?
- Session information passing between client and server

- Session information passing mechanisms - Cookies, Rewriting
- How to destroy a session

## **5. JSPs**

- Introduction to JSP and need for JSPs
- Basic HTML tags
- JSP Lifecycle

## **6. JSP Elements**

- Scriptlets
- Expressions
- Declarations
- Significance of above elements and fitment into the JSP Lifecycle
- What are Directives in JSP?
- Page Directive
- Include Directives
- Taglib Directive

## **7. JSP Tag library**

- JSP Standard Actions
- Expression Language
- JSTL basics and it's usage
- Need for Custom Tag Library
- Custom Tag Library implementation

# **Struts Framework (version 2.x)**

## **1. Basics of MVC**

- What is MVC?
- MVC Type1 and Type2 architecture
- Why Struts framework?
- Struts 1 overview
- Struts 1 and Struts 2 comparison

## **2. Struts 2 Architecture**

- Architecture Diagram explanation of following components:
- Components of Model, Views and Controller in Struts Framework
- Interceptors
- Model/Action classes
- Value Stack
- OGNL
- Introduction to configurations; framework and application architecture
- Declarative and Annotations configuration approaches

## **3. Struts 2 set up and first Action class**

- Download JAR files
- Struts 2 project build up and Configuration files
- To build Action class
- To intercept an HTTP request via Struts2 framework using Action class
- Defining data and business logic in Action class
- Preparing and Forwarding control to Views

## **4. Struts 2 Interceptors**

- What are Interceptors
- Responsibilities of an Interceptor
- Mechanism of Interceptor calling in Struts 2
- Defining Interceptors
- Defining Interceptor stacks
- Defining Custom Interceptors

## **5. Struts 2 Tag Library**

- Introduction to tag library of Struts 2 and it's usage

## **6. Struts 2 Validations**

- Validations using Validateable interface
- Workflow interceptor mechanism for validations

- Validations using Validateable interface
- Validation Framework introduction and architecture
- Validating user input with above two mechanisms

## **7. Struts 2 Tiles Frameworks**

- Introduction to Tiles in a page
- Struts2 Tiles framework introduction
- Defining tiles.xml file
- Configuring pages for tiles
- A complete Tiles example with Struts2

# Hibernate Framework (version 3.x)

## **1. Introduction**

- What is ORM principle?
- Why ORM?
- ORM implementations

## **2. Hibernate Architecture**

- Introduction to Hibernate
- Hibernate Architecture
- What are Persistent classes?

## **3. Hibernate CRUD**

- Setting up Hibernate project
- Configuring all JARs and XML files
- Setting up connection to DB using Hibernate
- Performing basic CRUD operations using Hibernate API
- Object Identity; Generator type classes
- Using SQL with Hibernate
- Using HQL
- Using Criteria queries

## **4. Mapping Collections and Associations**

- To define sets, maps, lists in Hibernate
- Association Mappings:
  1. One to one
  2. One to many
  3. Many to one
  4. Many to many
- Hibernate Caching
- What is caching?
- What are the types of caching in Hibernate?
- Explanation of various caching mechanisms in Hibernate

## **5. Using Hibernate Annotations**

- Sample example of using Hibernate Annotations

# **Spring Framework (version 3.x)**

## **1. Introduction to spring**

- What is Spring?
- Spring Architecture explanation and all its components

## **2. Introduction to all modules of Spring**

- Spring Bean Factory
- Spring Application Context
- Spring DI
- Spring Integration; Spring messaging, Spring JMS
- Spring MVC
- Spring DAO

## **3. Setting up spring**

- Setting up of Spring framework
- Download JARs

- Configure XML files

#### **4. Dependency Injection**

- What is Dependency Injection?
- How is it implemented using Spring Framework?
- Bean Wiring mechanisms in Spring

#### **5. Spring AOP**

- What is Spring AOP?
- Implementation of Spring AOP

## **Spring Boot Framework (Version 2.x)**

### **1. Introduction**

- Spring Boot Introduction
- Spring Boot Version
- Spring vs Spring Boot vs Spring MVC
- Spring Boot Architecture

### **2. Creating Project**

- Spring Initializr
- Download & Install STS IDE
- Spring Boot Example
- Spring Boot CLI
- Spring Boot Example-STs

### **3. Project Components**

- Annotations
- Dependency Management
- Application Properties
- Starters
- Starter Parent
- Starter Web

- Starter Data JPA
- Starter Actuator
- Starter Test
- Devtools
- Multi Module Project
- Packaging
- Auto-Configuration

#### **4. Tool Suite**

- Hello World Example
- Project Deployment Using Tomcat

#### **5. Spring Boot AOP**

- What is AOP?
- AOP Before Advice
- AOP After Advice
- AOP Around Advice
- After Returning Advice
- After Throwing Advice

#### **6. Spring Boot Database**

- JPA
- JDBC
- H2 Database
- Crud Operations

#### **7. Spring Boot View**

- Thymeleaf View

#### **8. Spring Boot Caching**

- What is Caching?
- Cache Provider

- EhCaching

## **9. Spring Boot Misc**

- Run Spring Boot Application
- Changing Port
- Spring Boot Rest Example

## **Web Services: REST and SOAP**

- Logging Framework: Splunk, Log4J, SLF4j
- Version-control system + repository hosting service: Git + Github